

Thanks for Paypal Tips !!

3DIY Captain <3diycaptain@gmail.com>
To: David Malawey <dmalawey@gmail.com>

Mon, Sep 11, 2023 at 9:16 AM

Hello David,
Finding a European vendor for the lens is difficult, I can only find a Chinese one. Here is one line for example: [10 pcs 20mm LED Lens 120 Degree with White Holder for 1W 3W High Power LED Light 609876957274 | eBay](#)
Using a 120° means you get more light from the scene. I think it was easier to find lenses before... Otherwise, PMMA lenses remain an option for the lens (I don't know how many optical losses it can induce...)
The optical fiber is the key part, I have this seller: [Câble fibre optique 0,125/0,25/0,5/0,75/1/1,5/2/ 2,8/ 4 mm | eBay](#)
You can contact him too for advice, he really helped me a lot during the project. optotravel@gmail.com
Have a good day,

Le mer. 6 sept. 2023 à 19:51, David Malawey <dmalawey@gmail.com> a écrit :
Thanks Damien,

Can you share a hyperlink for a glass lens you are considering? (1 inch as you say). A european vendor is OK - I can try to source equivalent parts in the USA.

Also, do you have a vendor or hyperlink for your recommended fiber? All of the affordable options I can find are plastic, with no datasheet.

Cheers,

David M

On Tue, Sep 5, 2023 at 8:34 AM 3DIY Captain <3diycaptain@gmail.com> wrote:
Hello David,

Thank you for your comments, the sun collector is still an important project to me :)

There was the 2020 upgrade in 2020, using 3D printed parts and glass lenses.

Here are a couple of thoughts:

1°/ the sun moving causes a loss of optical alignment in the duo lens/fiber, I think there are 2 ways to deal with this.

- by design, meaning, the configuration of the collector does not move with time, and there is always light feeding a fiber (it can be several fiber being fed one by one as the sun moves)

- sun tracker: a solution I'd love to try one day. I think it needs to be low cost (no arduino or other µcontroller), I found this video which is really interesting: https://www.youtube.com/watch?v=wL9PcGu_xrA

2°/ using good fiber (silica base) is a must, don't use plastic one, their optical losses per meter is really too high

3°/ always think about fire hazard: if the device is overheating and bursts into flames, you want to be sure nothing will burn next to it. For me a way to mitigate this effect is to use several small lenses (around 1inch - to be confirmed), so the solar power is lower, but still, you're ok with light. Also, by design, using parts that won't start a fire (like metal, glass...)

4°/ good point for the lens, I think using transparent filament is possible to make a thin lens (fresnel lens), curious to see if it will work.

5°/ Interesting fact, the colors at the output of the fiber depends on your target, for example aiming at trees will give you more green light

Really interested to see where you will go, please let me know and don't hesitate if I can help.

Cheers,

Damien

Le lun. 4 sept. 2023 à 17:11, David Malawey <dmalawey@gmail.com> a écrit :
Hello Damien,

You posted a great video on a [Sunlight Collector design](#).

I think you upgraded it [here in 2020](#)

I want to try to make this. Can you share some more details? Such as key parts, (I cannot find low-cost glass lenses), maybe a 3D print model? I may be able to upgrade the 3D model to reduce the number of parts, and I will share with you what I build. I am not doing anything for profit, it's for my house - not many windows.

Thanks for distributing your knowledge in the youtube videos! Very few people build these and document them!

Cheers,

David M

On Mon, Sep 4, 2023 at 9:13 AM 3DIY Captain <3diycaptain@gmail.com> wrote:
Hi David,
Thank you very much for your Paypal Tips!! It's really appreciated.
Please let me know if I can help.
Regards,

